

Telco Customer Churn Dashboard Insights Report

Business Analytics Case Study

1. Introduction

- Customer churn happens when customers stop using a company's service. For a telecom business, this is important because every lost customer can reduce revenue, weaken customer growth, and increase the cost of finding new customers.
- This report explains the main story from the Telco Customer Churn dashboard. The goal is to help stakeholders understand which customers are leaving, what patterns are connected to their decision, and what the business can do to improve retention.
- The dashboard focuses on customer status, contract type, payment method, internet service, tenure group, tech support, online security, and monthly charges. These areas are useful because they show customer commitment, customer experience, and possible service concerns.
- The report is written in simple business language so that both technical and non-technical stakeholders can understand the findings and act on them.

2. Business Problem Statement

- The business needs to understand why customers are leaving and which customer groups are most at risk.
- The main business question is: What factors are linked to customer churn, and what actions can the business take to reduce customer loss?
- The dashboard shows that customer churn is not happening evenly across all customers. It is concentrated in specific groups, especially customers on month-to-month contracts, newer customers, customers using electronic check, fiber optic users, and customers without support services.
- If the business does not address these areas, it may continue losing customers and revenue. If the business responds early, it can improve customer satisfaction, increase loyalty, and protect long-term revenue.

3. Dataset Description

- The dataset contains 7,043 customer records and 21 columns. Each row represents one customer.
- The Churn column shows whether a customer left the business or stayed. Customers marked as churned are those who left, while retained customers are those who stayed.
- The dataset includes customer information such as gender, senior citizen status, partner status, dependent status, tenure, phone service, internet service, online security, tech support, contract type, payment method, monthly charges, total charges, and churn status.
- The dashboard highlights 7,043 total customers, 1,869 churned customers, and 5,174 retained customers.
- The churn rate is 26.54%, meaning about one out of every four customers has left the business.
- The average monthly charge for churned customers is 74.44, while retained customers have a lower average monthly charge of 61.27.
- Churned customers have an average tenure of 17.98 months, while retained customers have an average tenure of 37.57 months.

4. Data Analysis Process

- The dataset was reviewed to understand the customer information available and how each field could help explain churn.
- Customer status was separated into two groups: customers who stayed and customers who left.
- Key measures were created to summarize the customer base, including total customers, churned customers, retained customers, churn rate, average monthly charge of churned customers, and average tenure.
- Customers were grouped by contract type to see whether customers with short or long contracts behaved differently.
- Customers were grouped by tenure to understand whether new customers or long-term customers were more likely to leave.
- Payment method was reviewed to check whether a particular payment option was linked to higher customer loss.
- Internet service type was reviewed to understand whether fiber optic, DSL, or no internet service had stronger churn patterns.
- Tech support and online security were reviewed because they show whether customers had access to support and protection services.
- The final dashboard was built to make the findings easy to understand using cards, donut charts, and bar charts.

5. Key Findings

- The overall churn rate is 26.54%. This means 1,869 out of 7,043 customers left the business. This is a major customer retention issue and should be monitored closely.
- Month-to-month contract customers are leaving the most. The dashboard shows 1,655 churned customers from this contract group, compared with 166 from one-year contracts and 48 from two-year contracts.
- New customers are leaving faster than long-term customers. The 0-12 month group recorded 1,037 churned customers, which is the highest tenure group on the dashboard.
- Churn reduces as customers stay longer with the business. The 61-72 month group recorded only 93 churned customers, showing that long-term customers are more stable.
- Electronic check customers have the highest churn count by payment method, with 1,071 churned customers. This is much higher than mailed check, bank transfer, and credit card users.
- Fiber optic customers have the highest churn count by internet service, with 1,297 churned customers. DSL recorded 459, while customers with no internet service recorded 113.
- Customers without tech support are leaving more. The dashboard shows 1,446 churned customers who did not have tech support.
- Customers without online security are also leaving more. The dashboard shows 1,461 churned customers who did not have online security.
- Churned customers pay more on average but stay for a shorter period. Their average monthly charge is 74.44, while their average tenure is only 17.98 months.

6. Business Insights

- The main story from the dashboard is that many customers are leaving before they build a strong relationship with the business.
- Customers on month-to-month contracts are the biggest risk group because they can leave easily. They do not have the same level of commitment as customers on one-year or two-year contracts.
- The first 12 months are very important. Since the highest number of churned customers are in the 0-12 month group, the business should treat the first year as a critical period for customer care and follow-up.
- Higher monthly charges may be creating pressure for customers. Churned customers have a higher average monthly bill than retained customers, which suggests that some customers may not feel they are getting enough value for what they pay.
- Fiber optic customers need special attention. They form the largest churn group by internet service, which may point to concerns around price, service quality, internet reliability, or customer expectations.
- Customers using electronic check should be reviewed closely. Their churn count is much higher than other payment groups, which may show that this customer group is less stable or may be facing payment-related issues.
- Customers without tech support are more likely to leave. This suggests that customers who cannot easily get help when they have service issues may become frustrated and cancel their service.
- Customers without online security are also more likely to leave. This may show that customers with fewer service features feel less connected to the business and can leave more easily.
- Longer-tenure customers appear more loyal. Customers who have stayed longer are less likely to leave, which means improving early customer experience can help create long-term customers.

7. Recommendations

- Create retention offers for month-to-month customers. Offer simple discounts, loyalty rewards, or added benefits to encourage them to move to one-year or two-year contracts.
- Improve the first-year customer experience. New customers should receive welcome calls, service guides, follow-up messages, and quick support during their first 12 months.
- Review pricing for customers with high monthly charges. The business should check whether these customers are receiving enough value and consider personalized offers for high-risk customers.
- Pay close attention to fiber optic customers. The business should collect feedback from these customers and check if the main concerns are price, speed, service reliability, or support response time.
- Encourage more stable payment methods. Customers using electronic check should be encouraged to use automatic bank transfer or credit card payment where possible.
- Promote tech support as a valuable service. Customers without tech support should be educated on how the service can help them resolve problems faster.
- Promote online security as part of customer protection. The business can offer simple bundles that include online security and support services.
- Create a churn watchlist. This list should focus on customers who are new, on month-to-month contracts, paying high monthly charges, using fiber optic, and missing tech support or online security.
- Train customer support teams to focus on early warning signs. If customers complain often, delay payments, or ask about cancellation, they should receive quick attention.
- Track churn monthly. Management should monitor churn rate, churned customers, new customer churn, and churn by contract type every month to measure improvement.

8. Conclusion

- The Telco Customer Churn dashboard shows that 1,869 customers left out of 7,043 customers, giving a churn rate of 26.54%. This means customer loss is a serious issue for the business.
- The strongest churn signals are month-to-month contracts, short tenure, electronic check payment, fiber optic internet service, lack of tech support, lack of online security, and higher monthly charges.
- The business should focus on keeping customers during their first year, encouraging longer contracts, improving support services, reviewing price value, and paying close attention to high-risk customer groups.
- Reducing churn will require consistent customer care, better service value, stronger support, and early action before customers decide to leave.
- If the business acts on these insights, it can retain more customers, protect revenue, and build stronger long-term customer relationships.